

HWK 4: Interactive Visualization

Two Creative Clocks for the Gym Treadmill

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GitHub Repo: github.com/Xinyi-Christie-Dong/Christie-imt-561-homeworks

Live Site: xinyi-christie-dong.github.io/Christie-imt-561-homeworks

Context of Use

Personal interest:

I go to the gym regularly and find treadmill sessions mentally draining — the built-in timer just counts up, giving no sense of what that time means. I wanted to design a clock that makes the same effort feel rewarding and even fun.



Who & Where

Gym-goers on a treadmill. Displayed on a screen mounted in front of the machine, replacing the default digital timer.



Problem

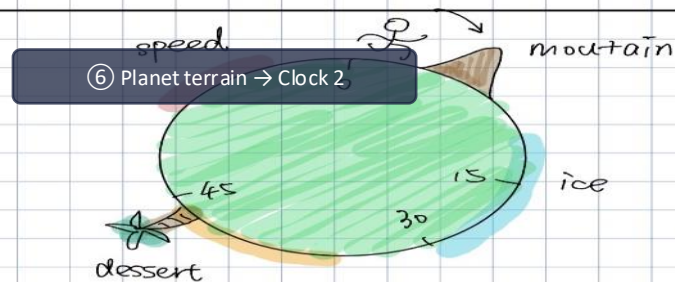
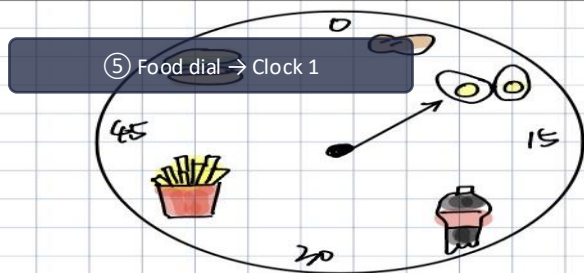
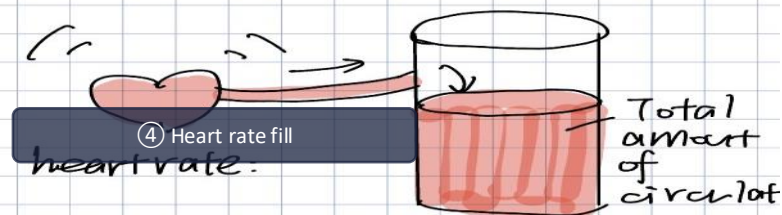
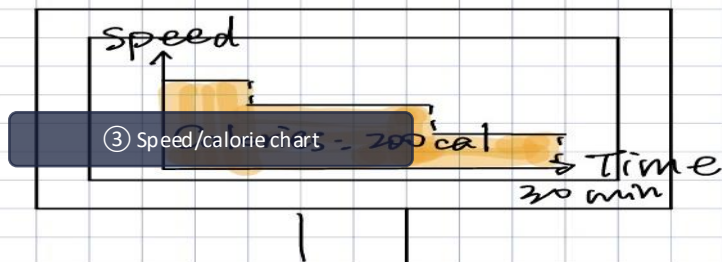
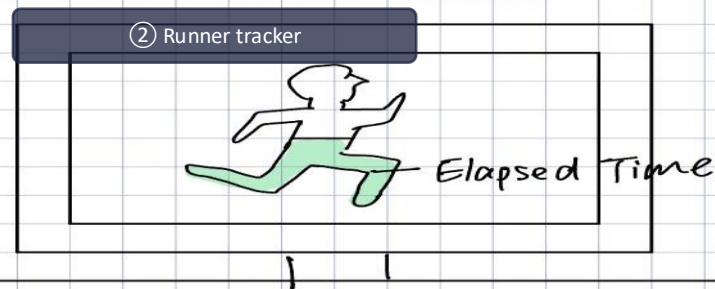
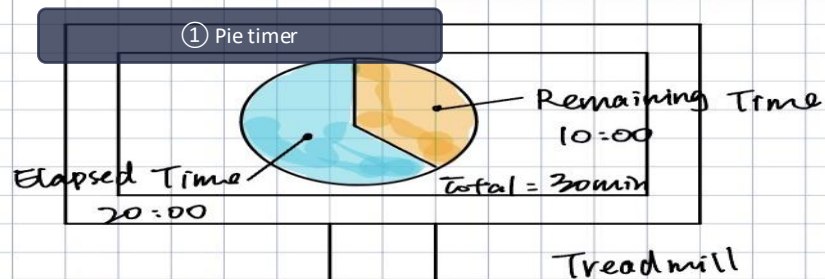
Treadmill sessions feel long and monotonous. A generic timer gives no sense of achievement — just numbers counting up.



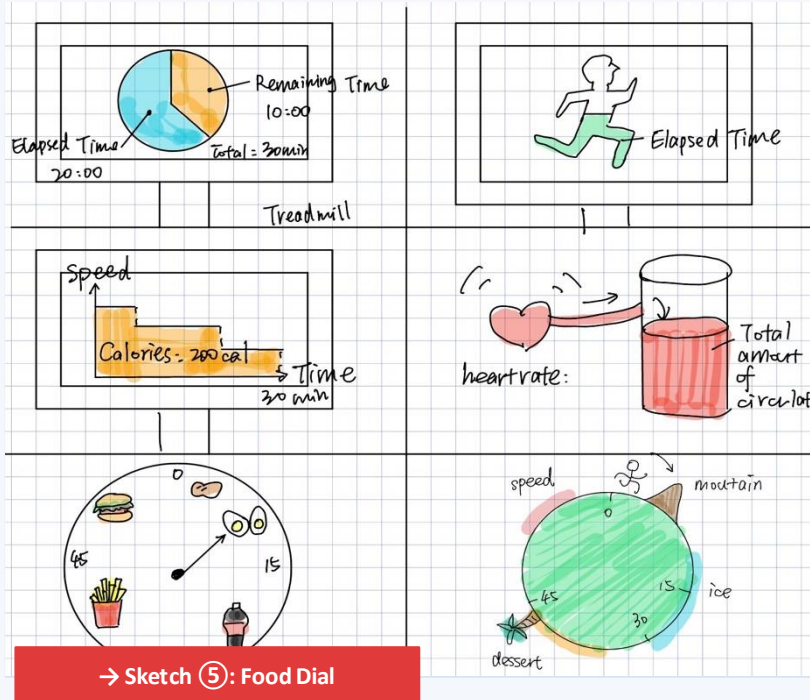
Solution

Two clocks that make time meaningful: one through calories as food earned, one through a game world where terrain warps time itself.

10 Hand-Drawn Sketches



Clock 1 — Treadmill Calorie Clock



Food icons at calorie-equivalent positions (~8 kcal/min) — turns time into a motivational food goal



Only the food the minute hand reaches activates; others gray — uses pre-attentive processing



Three time channels: minute hand + second hand + digital readout for glancing and precision



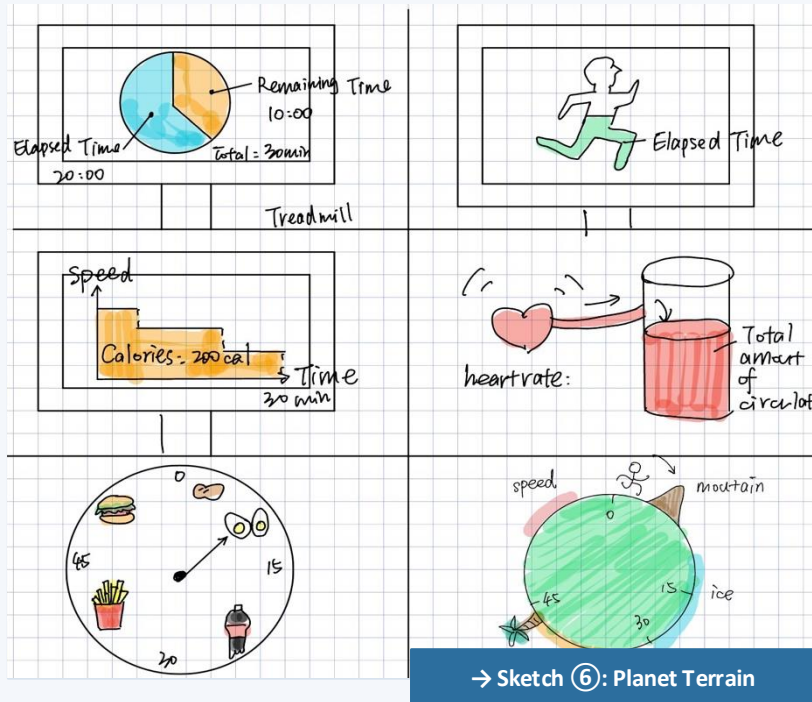
Laps Burned panel counts each food earned across laps — motivation builds over time



Feedback received & applied:

- Food icons overlapped tick marks → **moved outside ring**
- Dark background felt heavy → **switched to light mode**
- Calorie labels too small → **enlarged + bold on active food**

Clock 2 — Mario Planet Treadmill Clock



Non-linear ring: terrain zones warp arc size — 30 min $\neq$ halfway.
Encodes subjective effort, not just time



5 zones: Plain (1 $\times$), Mountain (0.5 $\times$), Ice (2 $\times$), Desert (0.7 $\times$),
Monster (3 $\times$)



Monster zone: pixel-art Goomba chases Mario — creates narrative
tension that resets each lap



Click to shuffle zones — same 60 min looks completely different
with different effort layout



Feedback received & applied:

- Dark space background felt disconnected → **light sky gradient**
- Gap between ring and planet body → **shrunk ring to remove gap**
- Boost zone felt generic → **replaced with Goomba monster chase mechanic**

Why the Two Designs Are Structurally Different

Clock 1 — Calorie Clock		Clock 2 — Mario Planet Clock	
Shape	Circular dial — classic clock form	Shape	Circular planet — non-clock form
Time encoding	Linear — minutes evenly spaced	Time encoding	Non-linear — zones warp the ring
Visual variable	Position (hand) + color (active food)	Visual variable	Arc length + character motion
Interaction	Space to start/pause; R to reset	Interaction	Click to shuffle terrain layout
Narrative	Static icons — you run toward food	Narrative	Dynamic — Goomba chases Mario
Key question	What have I burned so far?	Key question	How does effort warp time?

Links



GitHub Repository

<https://github.com/Xinyi-Christie-Dong/Christie-imt-561-homeworks>



Deployed Site

<https://xinyi-christie-dong.github.io/Christie-imt-561-homeworks/?tab=tab1>